



SIMATS ENGINEERING

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Course Code: MLA0408

Slot: C

Course Name: Deep Learning

Course Faculty: Dr. Sudhagar D

Project Title: SMART OBJECT RECOGNITION USING IMAGE ENHANCEMENT AND FEATURE INTELLIGENCE

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

Mevalurkuppam, Tamil Nadu, India
22°98'40"N, 80°17'13"E
Thursday, 13/08/2026 11:25 AM GMT +05:30

SYSTEM ARCHITECTURE
Smart Object Recognition using Image Enhancement and Feature Intelligence

1. INPUT IMAGE
2. IMAGE ENHANCEMENT & PREPROCESSING
3. HARRIS CORNER DETECTION
4. SIFT FEATURE EXTRACTION & MATCHING
5. PCA OPTIMIZATION
6. OBJECT RECOGNITION (CLASSIFICATION)

OUTPUT: Object Car, Confidence: 95.8%

Project Description: (here you write what you did in this project (contribution) including Model Description

Smart Object Recognition Using Image Enhancement and Feature Intelligence is an intelligent computer vision system designed to identify and classify objects from digital images accurately and efficiently. The project combines image enhancement techniques with feature extraction and machine learning/deep learning methods to improve object recognition performance. The system first preprocesses the input image by removing noise, improving brightness and contrast, and enhancing important visual details. The enhanced image is then analyzed to extract significant features such as shape, texture, edges, and patterns. These features are processed using an intelligent recognition model to detect and classify the objects present in the image. The main objective of the project is to improve recognition accuracy, particularly when images have poor lighting, noise, low contrast, or unclear object details.

Student Signature

Guide Signature